

NIDS Session Report

AI-Based Network Intrusion Detection System

Session Summary

Start Time:	1750243737.5630639
Stop Time:	2025-06-18T16:20:10.856499
Uptime:	73 seconds
Packets Processed:	116
Anomalies Detected:	14
Packet Rate:	0 packets/sec

Protocol Statistics

TCP PACKETS:	85
UDP PACKETS:	25
ICMP PACKETS:	0
OTHER PACKETS:	6

Alerts (5)

Type: PORT_SCAN | Time: 2025-06-18T16:19:00.851395

Details: {'src_ip': '10.0.2.15', 'dst_ip': '18.214.59.207', 'dst_port': 443, 'protocol': 6}

Explanation: A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.

Type: ML_ANOMALY | Time: 2025-06-18T16:19:02.603581

Details: {'anomaly_score': -7.080437415740772, 'features': {'packet_count': 5, 'total_bytes': 295, 'avg_packet_length': np.float64(59.0), 'std_packet_length': np.float64(2.0), 'min_packet_length': 55, 'max_packet_length': 60, 'avg_payload_size': np.float64(2.6), 'std_payload_size': np.float64(1.7435595774162693), 'total_payload_size': 13, 'tcp_count': 5, 'udp_count': 0, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 1, 'tcp_ratio': 1.0, 'udp_ratio': 0.0, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 2, 'unique_dst_ports': 5, 'total_ports_accessed': 5, 'common_port_ratio': 0.2, 'high_port_ratio': 0.8, 'avg_dst_port': np.float64(41054.4), 'std_dst_port': np.float64(20320.71109582536), 'min_dst_port': 443, 'max_dst_port': 51716, 'unique_connections': 5, 'total_connection_attempts': 5, 'avg_packets_per_connection': 1.0, 'connection_diversity': 1.0, 'max_packets_per_connection': 1, 'unique_src_ips': 5, 'unique_dst_ips': 2, 'src_ip_diversity': 1.0, 'dst_ip_diversity': 0.4, 'private_src_ip_ratio': 0.2, 'private_dst_ip_ratio': 0.5, 'packet_rate': 3.3342581241014093, 'inter_arrival_time_mean': np.float64(0.37489598989486694), 'inter_arrival_time_std': np.float64(0.21614876207717196), 'burst_count': 1, 'burst_ratio': 0.25, 'payload_size_mean': np.float64(2.6), 'payload_size_std': np.float64(1.7435595774162693),

'zero_payload_ratio': 0.0, 'large_payload_ratio': 0.0, 'payload_size_entropy': np.float64(0.7219280948873623), 'syn_count': 3, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 5, 'urg_count': 0, 'syn_ratio': 0.6, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:19:02.603566', 'packet_count': 5, 'protocol_distribution': {'tcp': 5, 'udp': 0, 'icmp': 0}, 'connection_info': {'unique_connections': 5, 'unique_src_ips': 5, 'unique_dst_ips': 2}}

Explanation: The machine learning model detected traffic that is statistically unusual compared to normal patterns.

Type: DDOS_INDICATOR | Time: 2025-06-18T16:19:16.258523

Details: {'src_ip': '34.232.130.202', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.

Type: PORT_SCAN | Time: 2025-06-18T16:20:01.818059

Details: {'src_ip': '10.0.2.15', 'dst_ip': '18.214.59.207', 'dst_port': 443, 'protocol': 6}

Explanation: A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.

Type: ML_ANOMALY | Time: 2025-06-18T16:20:02.818574

Details: {'anomaly_score': -7.22925639294125, 'features': {'packet_count': 100, 'total_bytes': 23137, 'avg_packet_length': np.float64(231.37), 'std_packet_length': np.float64(402.43338964355337), 'min_packet_length': 42, 'max_packet_length': 1536, 'avg_payload_size': np.float64(177.39), 'std_payload_size': np.float64(405.25317753226807), 'total_payload_size': 17739, 'tcp_count': 70, 'udp_count': 24, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.7446808510638298, 'udp_ratio': 0.2553191489361702, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 14, 'unique_dst_ports': 18, 'total_ports_accessed': 18, 'common_port_ratio': 0.3404255319148936, 'high_port_ratio': 0.6595744680851063, 'avg_dst_port': np.float64(35513.51063829787), 'std_dst_port': np.float64(25541.851134928), 'min_dst_port': 53, 'max_dst_port': 64272, 'unique_connections': 17, 'total_connection_attempts': 94, 'avg_packets_per_connection': 5.529411764705882, 'connection_diversity': 0.17, 'max_packets_per_connection': 14, 'unique_src_ips': 10, 'unique_dst_ips': 9, 'src_ip_diversity': 0.1, 'dst_ip_diversity': 0.09, 'private_src_ip_ratio': 0.2, 'private_dst_ip_ratio': 0.2222222222222222, 'packet_rate': 2.123891303009895, 'inter_arrival_time_mean': np.float64(0.4755897859130243), 'inter_arrival_time_std': np.float64(1.4000144630789715), 'burst_count': 51, 'burst_ratio': 0.5151515151515151, 'payload_size_mean': np.float64(177.39), 'payload_size_std': np.float64(405.25317753226807), 'zero_payload_ratio': 0.18, 'large_payload_ratio': 0.1, 'payload_size_entropy': np.float64(1.1004680459013572), 'syn_count': 29, 'fin_count': 2, 'rst_count': 3, 'psh_count': 12, 'ack_count': 65, 'urg_count': 0, 'syn_ratio': 0.4142857142857143, 'fin_ratio': 0.02857142857142857, 'rst_ratio': 0.04285714285714286, 'psh_ratio': 0.17142857142857143, 'ack_ratio': 0.9285714285714286, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:20:02.818560', 'packet_count': 100, 'protocol_distribution': {'tcp': 70, 'udp': 24, 'icmp': 0}, 'connection_info': {'unique_connections': 17, 'unique_src_ips': 10, 'unique_dst_ips': 9}}

Explanation: The machine learning model detected traffic that is statistically unusual compared to normal patterns.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
port_scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.
ddos_indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
unusual_protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ml_anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.